

HAFIZ MUHAMMAD

AI & Backend Engineer • Python • FastAPI • Django • LangChain & LangGraph • React
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PROFESSIONAL SUMMARY

AI & Backend Engineer currently working at PureLogics, Lahore — building production-grade Python web applications, RESTful APIs, and AI-powered systems. Specializes in LangChain and LangGraph for building intelligent, multi-step AI agents and agentic workflows. Experienced with RAG pipelines, vector databases, real-time AI voice systems, and full-stack web development using Django, FastAPI, and React.

TECHNICAL SKILLS

AI & Agentic Systems: LangChain • LangGraph • RAG Pipelines • ChromaDB • Groq API • Ollama • Whisper AI • OpenCV

Backend: Python • FastAPI • Django • Flask

Frontend: React • JavaScript • HTML & CSS • WebSockets

Databases: MySQL • SQLAlchemy • ChromaDB

Dev Practices: OOP • MVC Architecture • RESTful APIs • Git & GitHub • Linux Deployment

Certifications: Databases & SQL (IBM) • OOP in Java • Design Patterns • Linux Management

PROFESSIONAL EXPERIENCE

AI & Backend Engineering | PureLogics, Lahore | July 2025 – Present

- Build scalable Python web apps using Django and FastAPI with MVC architecture and clean coding standards in a team environment.
- Design RESTful API endpoints and collaborate with frontend developers to integrate server-side logic with user-facing features.
- Independently build AI service modules and LangChain/LangGraph-powered agentic pipelines; participate in code reviews.
- Manage MySQL databases, deploy applications on Linux servers, and use Git for version control and team collaboration.

PROJECTS

Full source code available at: hafizmuhammadshah42895.github.io/hafizmuhammad.dev

CallerAI – AI-Powered Cold Calling Platform — Full Stack Developer | 2025

An AI voice agent that handles sales calls for businesses — speaks naturally, knows the company's products, and logs every conversation with a summary and sentiment score.

- Built a real-time AI voice pipeline using **LangGraph** to orchestrate: Whisper AI (speech-to-text) → LLaMA 3.1 via Groq (AI thinking) → Cartesia Sonic-2 (voice output) — all streamed live over WebSockets.
- Used **LangChain + ChromaDB** to build a private RAG knowledge base — the AI only answers about the business's products and deflects off-topic questions.
- Integrated Telnx for real phone calls (PSTN) and a browser-based web call mode. Added automatic post-call analysis: summary, sentiment, and outcome label.

Stack: Python, FastAPI, LangChain, LangGraph, ChromaDB, Whisper AI, Groq, Telnx, WebSockets, MySQL, React, Git, Linux

Premium Matrimony – AI Matchmaking Platform — Full Stack Developer | 2025

An online marriage platform for Pakistani and South Asian users with AI-powered match suggestions, real-time chat, OTP authentication, and a full admin panel.

- Built AI match suggestions using **LangChain + ChromaDB vector embeddings** — the system finds the top 5 most compatible partners and explains in plain English why they match.
- Added a feedback learning loop: liking or skipping matches updates the user's preference profile for smarter future suggestions.
- Built real-time messaging (WebSockets), OTP email verification, JWT auth, and a full admin panel for user management and CNIC verification.

Stack: Python, Django, FastAPI, LangChain, ChromaDB, WebSockets, MySQL, React, JWT, REST APIs, Git, Linux

Movers – Logistics Booking Platform — Lead Developer | 2025

A full-stack app where customers book moving services and pay online. Built end-to-end from database design to Stripe payment integration.

- FastAPI backend with MVC architecture, RESTful APIs, Stripe payments, MySQL schema design, and JavaScript frontend integration.

Stack: Python, FastAPI, Django ORM, MySQL, JavaScript, Stripe, REST APIs, Git

Gourmet Haven – Restaurant Chatbot App — Backend Developer | 2025

A restaurant web app with a conversational AI assistant that answers questions about dishes, ingredients, and recommendations.

- Flask MVC backend with clean OOP Python, RESTful API design, and modular code using software design patterns.

Stack: Python, Flask, MVC, OOP, RESTful APIs, JavaScript, Git

AI-Powered Video Assistant — [Full Stack Developer](#) | 2025

Upload a video, and the app transcribes what was said using AI — then lets you ask questions about the video content.

- Flask MVC app with Whisper AI for transcription, RESTful APIs, MySQL database, and full Linux server deployment.

Stack: Python, Flask, MVC, Whisper AI, HTML, CSS, JavaScript, MySQL, Git, Linux

EDUCATION & CERTIFICATIONS

B.Sc. Computer Science | [Riphah International University](#)

Certifications: Databases & SQL for Data Science (IBM) • OOP in Java (Oracle) • Design Patterns (FreeCodeCamp) • Linux Management (Linux Foundation)